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Exercise 7

$P(A) = 0.5$ $P(WS|A) = 3/5$ $P(WS|B) = 1/3$ $P(WS|C) = 3/8$

$P(B) = 0.3$ $P(D|A) = 1/10$ $P(D|B) = 1/2$ $P(D|C) = 1/4$

$P(C) = 0.2$ $P(CM|A) = 3/10$ $P(CM|B) = 1/6$ $P(CM|C) = 3/8$

i) $$P(A|CM) = \frac{P(CM|A)P(A)}{P(CM|A)P(A) + P(CM|B)P(B) + P(CM|C)P(C)}$$

$$= \frac{(3/10)(0.5)}{(3/10)(0.5) + (1/6)(0.3) + (3/8)(0.2)}$$

$$= 6/11$$

ii) $$P(C|WS) = \frac{P(WS|C)P(C)}{P(WS|A)P(A) + P(WS|B)P(B) + P(WS|C)P(C)}$$

$$= \frac{(3/8)(0.2)}{(3/5)(0.5) + (1/3)(0.3) + (3/8)(0.2)}$$

$$= 3/19$$

iii) $$P(B|D) = \frac{P(D|B)P(B)}{P(D|A)P(A) + P(D|B)P(B) + P(D|C)P(C)}$$

$$= \frac{(1/2)(0.3)}{(1/10)(0.5) + (1/2)(0.3) + (1/4)(0.2)}$$

$$= 3/5$$

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Exercise 8

$P(M) = 0.51$

(a) $P(\bar{M}) = 1 - P(M)$

$= 1 - 0.51$

$= 0.49$

(b) $P(R|M) = 0.095$ $P(R|\bar{M}) = 0.017$

i) $P(R|M) = 0.095$

ii) $$P(M|R) = \frac{P(R|M)P(M)}{P(R|M)P(M) + P(R|\bar{M})P(\bar{M})}$$

$$= \frac{(0.095)(0.51)}{(0.095)(0.51) + (0.017)(0.49)}$$

$= 0.853$

iii)

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Exercise 9

P = Prison $P(P) = 0.45$

G = Guilty $P(G|P) = 0.4$

$P(G|P') = 0.55$

i) $P(P') = 1 - P(P)$

$= 1 - 0.45$

$= 0.55$

ii) $$P(P|G) = \frac{P(G|P)\,P(P)}{P(G|P)\,P(P) + P(G|P')\,P(P')}$$

$$= \frac{(0.4)(0.45)}{(0.4)(0.45) + (0.55)(0.55)}$$

$= 0.3731$

iii) $$P(P'|G) = \frac{P(G|P')\,P(P')}{P(G|P)\,P(P) + P(G|P')\,P(P')}$$

$$= \frac{(0.55)(0.55)}{(0.4)(0.45) + (0.55)(0.55)}$$

$= 0.6269$

iv) $P(G) = P(G|P)P(P) + P(G|P')P(P')$

$= (0.4)(0.45) + (0.55)(0.55)$

$= 0.4825$